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SARDAR PATEL UNIVERSITY

Second Year BCA (SEM-III) Examination

US03BCA03: Advanced Data and File Structures [2020]

DATE: 13/12/2020, Thursday **TIME:** 10:00 am to 1:00 p.m **Total Marks: 70**

Note: 1. All the questions are compulsory. 2. Figures to the right indicate marks.
3. Start a new question from a new page.

Q.1 Choose the correct answer:

[10]

- a. An array is a _____ data structure.
1. Unordered 2. Non-composite 3. Composite 4. Heterogeneous
- b. Tree is a _____ data structure.
1. Linear 2. Non-Linear 3. Variable 4. Non-subscript
- c. _____ is a rank of hierarchy of a tree.
1. Degree 2. Level 3. Height 4. Index
- d. A Binary tree has at most _____ child.
1. One 2. Two 3. Three 4. None of these
- e. A node whose indegree is 0 is called _____.
1. Source node 2. Self loop 3. Single node 4. Cycle
- f. A directed edge is known as _____.
1. Segment 2. Double edge 3. Arc 4. Arrow
- g. There are at most _____ passes required to perform bubble sort.
1. $n - 1$ 2. n 3. $n + 1$ 4. $n / 2$
- h. The process of finding the data from its data structure is called _____.
1. Searching 2. Deletion 3. Sorting 4. None of these
- i. In _____ file stored on tape, it is not possible to rewrite records.
1. Serial 2. Sequential 3. Direct 4. None of these
- j. CDC stands for _____.
1. Control Data Cycle 2. Control Data Corporation
3. Control Data Centre 4. None of these

Q.2 Answer any 10(ten) from the following:

[20]

- a. Define Array with example.
- b. Define tree with an example.
- c. Define leaf of a tree with example.
- d. Draw the Binary Tree for $(A + B) + C * (E / F)$
- e. Define Directed Graph.
- f. Define parallel edges of a Graph.
- g. Define searching. Also list the searching techniques.
- h. Define Binary search.
- i. List the applications of Searching.
- j. Define: Record, Key.
- k. Define: Address space, Key space.
- l. Explain in brief index area.

- Q.3
a. Explain 1-D array with declaration and initialization. [05]
b. Explain applications of an array. [05]
OR
- Q.3
a. Explain 2-D array with declaration and initialization. [05]
b. Explain applications of tree. [05]
- Q.4
a. What are the types of traversal of Binary tree? Explain Preorder traversal with an example. [05]
b. Explain the concepts of Undirected graph, Connected Graph, Sink Node with an example and diagram. [05]
OR
- Q.4
a. Explain Lexically ordered tree. [05]
b. Explain the insertion of a node in a binary tree with example. [05]
- Q.5
a. What is sorting? List sorting techniques and explain any one of them (for ascending sort) with algorithm using the following data: [10]
55, 22, 77, 92, 12
OR
- Q.5
a. What is searching? List and explain any one of them with algorithm using the following data: [10]
55, 22, 77, 92, 12
Element to be searched : 12
- Q.6
a. Write a short note on Multiple buffering. [05]
b. Explain the structure of index sequential file supported by IBM. [05]
OR
- Q.6
a. Write a detailed note on processing of Direct file. [05]
b. Explain the structure of index sequential file supported by CDC. [05]

*****X*****